

Jaume **Ivars Grimalt**

Software Engineer · Backend, Infrastructure & Full-Stack

Currently in Valencia, Spain | EU citizen | Open to relocation to Zurich, Munich, London

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Profile

I build and run production systems end to end. At Mycrospace I was the sole engineer: I built the backend, the async inference pipeline, the cloud infrastructure on GCP and the web and mobile apps, and managed an intern. I also trained the computer vision models the platform serves, so I'm comfortable where software meets ML.

Technical Skills

Languages	TypeScript, Java, Python
Backend	NestJS, Spring Boot, Prisma, PostgreSQL, MongoDB, Redis/BullMQ, SSE/WebSocket, Stripe
Frontend & Mobile	Next.js, React, React Native/Expo
Cloud & DevOps	GCP (Cloud Run, Cloud SQL, GCS), Docker, CI/CD, multi-GPU HPC (SLURM)
ML	PyTorch, ONNX, multi-GPU training (DDP, AMP)

Experience

Mycrospace — *Valencia, Spain*

Co-Founder & CTO — *Jan 2024 – Oct 2026*

- Computer vision product for microbiology labs, from Master's thesis to company; tested in **5 labs**; 1st Prize IdeasUPV (2024). Sole engineer; managed an intern.
- **Backend.** NestJS/Prisma/PostgreSQL API and MongoDB; async inference jobs with BullMQ/Redis; signed-URL uploads to GCS; real-time updates over SSE/WebSocket.
- **Infra.** Deployed on GCP (Cloud Run, Cloud SQL, GCS) with Docker and CI/CD; ONNX models in production.
- **Frontend.** Next.js web app and React Native/Expo mobile app; Stripe payments.
- **ML.** Trained the detection model behind the product from scratch (**12,500+** GPU-hours on the EuroHPC Leonardo supercomputer, multi-GPU PyTorch on SLURM).

MIALAB: Medical Imaging Analysis Laboratory, UPV — *Valencia, Spain*

Research Engineer (full-time) — *May 2025 – Jan 2026*

- Profiled the volBrain pipeline (**700k+** volumes) from NIfTI ingest to voxel prediction for inference cost; volumetric brain MRI segmentation with nnU-Net.

Tapstar — *Valencia, Spain*

Full-Stack Engineer (freelance) — *Jan 2025 – May 2025*

- QR generation, back office and frontend for QR cards and stands; scan analytics and per-employee ratings.

Vesta-Z — *Valencia, Spain*

Data Engineer — *Jul 2023 – Feb 2025*

- Full-stack cloud features; custom schema for cross-matching items across providers; LLM customer-support bot.

Earlier experience: Neurocatching, Data Scientist Intern, 2023; Sciling, Backend Developer Intern, 2022.

Education

M.Sc. Artificial Intelligence, Pattern Recognition & Digital Imaging — *Valencia, Spain*

UPV · GPA 8.7/10 · Thesis Honours (became Mycrospace) — *2023 – 2024*

Technology Management — *Valencia / Munich*

CDTM (TUM/LMU) · 16 selected — *2024*

B.Sc. Computer Engineering — *Valencia, Spain*

UPV · GPA 8.3/10 · Honours in ML and Statistics — *2019 – 2023*

Publications & Research

- *Auto-N2N: Self-supervised MRI Denoising using Synthetic Pair Generation and Mixture of Experts.* Preprint, 2026.
- *Detecting Cytopathic Effect in Cell Culture with Computer Vision.* Under review.
- *What Extra Euler Steps Reveal About Residual Networks: Feature Dynamics in Weight-Tied and Untied Stages.* Manuscript in preparation.

Languages: Spanish (native), Catalan (native), English (C1)